ExtraCredit 19 April 2026

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Question 1

We found Approximate Distribution of Posteior Distribution for

$$\lambda | X_1 = 3, X_2 = 5, X_3 = 9$$

likely chi square distribution. but we can't observe exact posterior distrubution because they simulation many time

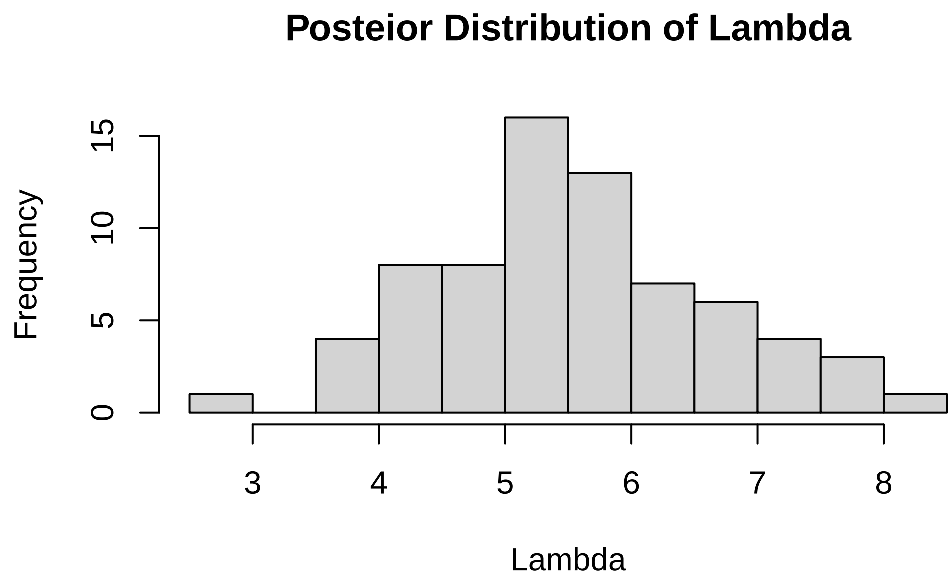
```
## Sample Distribution
collection <- c()

for (i in 1:30000){
  ## Prior Distribution
  lambda <- rchisq(n=1,df=5)
  ## Sampling (Simulate) and Sorting
  X <- sort(rpois(n = 3, lambda = lambda))
  out <- c(lambda, X)
  collection <- rbind(collection,out)
}

colnames(collection) <- c('lambda','x1','x2','x3')

# Create Index when value match observe data
idx <- (collection[,2]==3)&(collection[,3]==5)&(collection[,4]==9)

hist(collection[idx,1]
      ,main = 'Posteior Distribution of Lambda'
      ,xlab = 'Lambda'
      )
```



Question 2

99% Credible Interval of Lambda is

```
lambda_lower <- quantile(collection[idx,1],0.005)
lambda_upper <- quantile(collection[idx,1],1-0.005)

print(lambda_lower)
```

```
    0.5%
3.052908
```

```
print(lambda_upper)
```

```
  99.5%
8.14425
```

Question 3

We found

$$P(\lambda \leq 6|PosteriorDistribution) > P(\lambda > 6|PosteriorDistribution)$$

Then we fail to reject null hypothesis.

```
post_lambda <- collection[idx,1]

# Calculate Prob for Null Hypothesis
idx_h0 <- (post_lambda<=7)
count_h0 <- length(post_lambda[idx_h0])
prob_h0 <- count_h0/length(post_lambda)

print(prob_h0)
```

```
[1] 0.8873239
```

```
# Calculate Prob for Null Hypothesis
print(1-prob_h0)
```

```
[1] 0.1126761
```